

DEPLOYING 802.11N

Technology overview and checklist for network administrators who are planning to deploy 802.11n wireless networking



■ RAHUL KULKARNI

The latest buzzword in the Wi-Fi technology is the 802.11n Draft 2.0. 802.11n is an enhancement to the IEEE 802.11 wireless network standard that includes many new features to increase transmission speeds, range and reliability compared to 802.11a/b/g. These enhancements ideally translate to 300 Mbps of raw data throughput and/or double the range compared to current 802.11a/b/g technology.

IEEE 802.11n technology overview

The current wireless solutions operate in the 2.4GHz radio frequency band

(802.11g and 802.11b) or the 5GHz radio band (802.11a). 802.11n can operate on both 2.4GHz as well as 5GHz bands. It provides backwards compatibility by supporting simultaneous 802.11a and 802.11n (802.11n/a) at 5 GHz and/or simultaneous 802.11b, 802.11g and

802.11n (802.11n/b/g) at 2.4 GHz.

The 802.11n standard makes use of several innovative technologies to improve the range, throughput and reliability of wireless local-area networks (LANs). The three primary innovations are multiple-input multiple-output (MIMO) technology, packet aggregation and channel bonding (40MHz channels). Together, these techniques allow 802.11n solutions to achieve approximately fivefold increase in performance over current 802.11a/b/g networks.

Multiple-input multiple-output. On the transmitter side, MIMO allows multiple data streams to use the same frequency but over different spatial channels via spatial division multiplexing (SDM). This increases both the transmission power and data that can be sent over air. On the receiver side, MIMO can combine signals from multiple paths to improve reception quality.

While previous wireless technologies had problems dealing with multipath signal reflections and attenuation, MIMO actually uses these reflections to increase the range and reliability over the wireless coverage area.

Channel bonding. The 802.11a/b/g

TABLE I
Comparison of Various Wireless LAN Standards

IEEE Standards	802.11b	802.11g	802.11a	802.11n
Standard approved	Jul-99	Jun-03	Jul-99	Draft-N
Max. data rate	11 Mbps	54 Mbps	54 Mbps	300 Mbps
Modulation	DSSS	OFDM	OFDM	SDM
Frequency	2.4 GHz	2.4 GHz	5 GHz	2.4 GHz or 5 GHz
No. of spatial streams	1	1	1	2
Channel width	20 MHz	20 MHz	20 MHz	20 MHz/40 MHz